

Gaze in the Crowd: Frustum-Aware Feature Aggregation for Robust Gaze Target Estimation

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Abstract

As a core capability for human intent inference, gaze target estimation remains challenging due to severe contextual interference in crowded scenes. Traditional multi-branch methods suffer from cumbersome architectures and heavy reliance on auxiliary modalities. Meanwhile, recent single-branch models based on Vision Foundation Models (VFM) blindly exploit only the deepest features. Especially in crowded environments, this over-reliance on deep representations causes VFMs to struggle with semantic confusion and lose the high-resolution geometric details essential for accurate gaze tracking. To address this, we present GazeSpot, which unleashes multi-layer foundation features to restore the geometric-semantic balance via two key contributions: First, the Geometry-Guided Spatial Focus (GGSF) module leverages a physical frustum prior to generate an explicit multiplicative mask. Acting like a spotlight, GGSF proactively filters background noise and mitigates contextual interference to resolve spatial focus. Second, the Scale-Aware Semantic Aggregation (SASA) module adaptively fuses shallow, occlusion-resistant textures with deep semantics based on the global context, effectively addressing the feature granularity selection problem. We curate a specialized Crowd evaluation subset to facilitate robust benchmarking under severe social interference. Extensive experiments on general public benchmarks and the Crowd subset demonstrate that our method achieves state-of-the-art performance. Code and pre-trained models will be released.

CCS Concepts

• **Computing methodologies** → **Scene understanding**; *Neural networks*; • **Human-centered computing** → Human computer interaction (HCI).

Keywords

Gaze Target Estimation, Vision Foundation Models, Spatial Attention, Multi-scale Feature Aggregation, Human Behavior Analysis

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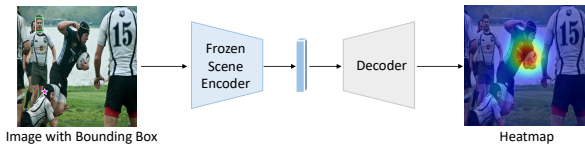
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(a) Single-Layer Semantic Bottleneck (Existing Methods)



(b) Geometry-Guided Multi-Scale Paradigm (Ours)

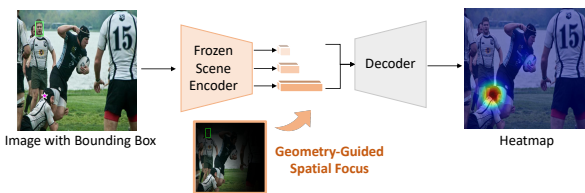


Figure 1: Motivation comparison. Existing methods (top) exploit only the deepest foundation features, trapping themselves in the "spatial-semantic dilemma" with diffuse activations and lost geometric details. In contrast, our proposed GazeSpot (bottom) introduces a geometry-guided multi-scale mechanism, enabling a paradigm shift from "blind semantic reliance" to "adaptive geometric-semantic synergy."

1 Introduction

Gaze target estimation aims to infer the final fixation point of a person within an image [27]. It serves as a core driving force for high-level semantic understanding in computer vision, such as human intent inference [11, 17, 23, 40], human-computer interaction (HCI) [25, 42], and social behavior analysis [9]. Given an image containing a person and their head bounding box, this task requires the model to not only precisely capture the 3D head pose [12, 15] and subtle eyeball deflections [10] but also possess a profound understanding of scene semantics [27]. This is crucial for inferring the physical interaction between the gaze direction and scene objects.

Traditional gaze estimation methods heavily rely on complex multi-branch architectures [3, 5, 16, 28, 30, 39, 44] (processing the head and scene independently) and auxiliary modalities like depth maps [1, 13, 15, 36]. Recently, inspired by the powerful generalization capabilities of Vision Foundation Models (VFMs), advanced methods like Gazelle [29] have pioneered a minimalist single-branch paradigm. These approaches discard cumbersome feature fusion and auxiliary inputs, instead feeding the entire image into a frozen VFM and directly utilizing its final-layer features for unified scene representation and gaze decoding. However, a straightforward question arises: *Are these single-layer minimalist paradigms sufficiently robust in complex, real-world crowded scenes?* To investigate this, we upgraded the backbone

of a baseline model from DINOv2 [26] to DINOv3 [32], aiming to leverage its stronger semantic representation capabilities. Counter-intuitively, the DINOv3-equipped model not only underperforms the DINOv2 baseline on the overall dataset, but the performance gap between them widens on the Crowd evaluation subset (detailed in subsection 4.5).

Through in-depth analysis, we discovered that the blind reliance on the extremely deep features of large models traps them in a severe **spatial-semantic dilemma**. As illustrated by the Principal Component Analysis (PCA) visualization in Figure 2, during the layer-by-layer forward pass of VFMs, the model acquires high semantic homogeneity as the network depth increases and the receptive field expands drastically. Compared to DINOv2, the deepest features of DINOv3 exhibit severe **feature over-smoothing** [41] in crowded regions. Gaze estimation is not merely a semantic classification task; it is highly sensitive to geometric and physical properties [43]. For instance, in highly dynamic multi-person interaction scenarios with occlusions (as shown in Figure 1), when a person looks at a complex tackling action below, the model heavily depends on the fine spatial structures provided by shallow or middle-layer features to accurately track the gaze direction. However, the global receptive field of a single deep feature leads to severe **contextual interference** and **semantic confusion**. Consequently, the geometric features of the target person become entangled with adjacent salient distractors. This irreversible loss of high-resolution geometric details causes models utilizing only the last-layer features to easily have their attention hijacked [35] in complex scenes, resulting in predictions that are diffuse, penetrate occlusions, or completely deviate from the target.

Motivated by this insight, to break the representation bottleneck imposed by a single deep layer, we propose an efficient single-branch gaze estimation framework named **GazeSpot**. This framework unleashes the potential of multi-layer foundation features for gaze inference without introducing significant computational overhead. Specifically, to address the aforementioned dilemma, we introduce two core designs to reconstruct the geometric-semantic balance.

First, deep networks are prone to losing relative spatial positions, leading to diffuse activation distributions. To counter this, we propose the **Geometry-Guided Spatial Focus (GGSF)** module. Abandoning the traditional practice of implicitly learning spatial positions, GGSF directly transforms the prior coordinates of the person's head into a physically meaningful relative distance field. Through a lightweight perceptron, GGSF maps this distance field into a spotlight-like frustum mask (Spotlight Mask). During the early stages of feature fusion, GGSF employs a multiplicative gating mechanism to act as a physical shield. It proactively filters out background activations outside the gaze cone, fundamentally cutting off contextual interference and resolving the "where to look" spatial focus problem.

Second, to overcome the limitations of feature over-smoothing in single-layer representations, we design the **Scale-Aware Semantic Aggregation (SASA)** mechanism. SASA performs global context perception across multi-layer features (shallow, middle, and deep) to dynamically compute attention weights for features at each scale. This mechanism acts as the model's cognitive engine, enabling it to adaptively route features based on image content. In crowded

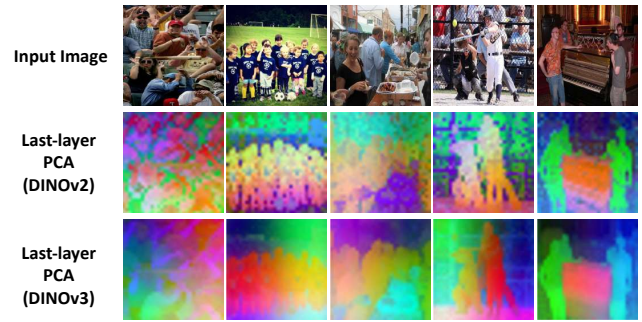


Figure 2: PCA visualization of the last-layer features from DINOv2 and DINOv3 in crowded scenes. While DINOv3 possesses stronger global semantic representations, it exhibits severe feature over-smoothing compared to DINOv2. This semantic homogeneity blurs the geometric boundaries between distinct individuals, causing severe contextual interference and the spatial-semantic dilemma in gaze estimation.

scenes, it intelligently invokes shallow textures to provide occlusion-resistant sharp edges, achieving optimal recombination with deep semantics. This effectively solves the feature granularity selection problem of "what to look at."

This dual design endows GazeSpot with both multi-scale semantic capacity and strict spatial geometric constraints. Furthermore, to accurately evaluate the model's anti-interference capability in complex social scenarios, we specifically curated a **Crowd evaluation subset**. Under a minimalist setup with no auxiliary modality inputs, our method not only demonstrates excellent performance on general benchmarks but also proves its robustness in highly challenging crowded scenes.

The core contributions of this paper are summarized as follows:

- **New Perspective.** We deeply reveal the "spatial-semantic dilemma" and contextual interference faced by single-branch VFM models in crowded scenes. We demonstrate the necessity of introducing multi-layer feature adaptive routing to restore geometric perception capabilities.
- **New Module.** We propose the Geometry-Guided Spatial Focus module. By constructing an explicit relative distance field and frustum isolation, it effectively cuts off the interference of background noise, significantly enhancing the model's spatial focus capability and interpretability.
- **New Mechanism.** We propose the Scale-Aware Semantic Aggregation mechanism, endowing the model with cross-scale feature routing capabilities to dynamically allocate shallow occlusion-resistant textures and deep global semantics based on the global context.
- **New State-of-the-Art.** We curated a Crowd evaluation subset specifically targeting multi-person interference. GazeSpot not only achieves new SOTA performance on the general GazeFollow and VideoAttentionTarget datasets but also exhibits remarkable robustness advantages on the Crowd subset.

2 Related Work

2.1 Gaze Target Estimation Paradigms

Early approaches to gaze target estimation established multi-branch architectures as the dominant paradigm. Recasens et al. [27] pioneered independent encoders for head crops and scene images, while Chong et al. [5] extended this framework to video sequences, introducing temporal modeling for dynamic scenes. Subsequent works incorporated multi-modal cues, with Fang et al. [10] and Gupta et al. [14] integrating depth and pose estimation to enhance spatial reasoning, while Tonini et al. [37] incorporated explicit object detection branches. However, these methods rely on complex cross-branch fusion and auxiliary models, resulting in parameter-heavy architectures sensitive to modality noise. Recently, Ryan et al. [29] and Song et al. [33] shifted toward single-branch foundation model paradigms using frozen DINOv2 encoders [26], significantly simplifying architectures while achieving competitive performance. Nevertheless, these approaches typically depend on single-layer deep representations, which may limit fine-grained spatial reasoning in complex scenes.

2.2 Representation Learning for Dense Prediction

While Vision Foundation Models such as DINOv2 provide powerful semantic representations, deep features exhibit inherent limitations in dense prediction tasks. Caron et al. [2] and Siméoni et al. [31] observed that self-supervised transformers tend toward feature over-smoothing, progressively losing fine-grained spatial details as the receptive field expands. For gaze estimation specifically, Kellnhofer et al. [19] emphasized that accurate localization relies on high-resolution geometric cues such as subtle head pose variations, while Chong et al. [5] noted that distinguishing targets among spatially proximal distractors requires preserving fine spatial structures. These observations suggest that aggregating hierarchical representations [21] is necessary to balance semantic abstraction and geometric precision, rendering single-layer representations insufficient for crowded scenarios.

2.3 Spatial Modeling and Attention Constraints

Contemporary gaze estimation systems predominantly employ attention mechanisms for spatial modeling. Dosovitskiy et al. [8] demonstrated that Transformers capture long-range dependencies through self-attention, yet such mechanisms provide only soft constraints on spatial relationships. Tafasca et al. [35] observed that in crowded scenes, the indiscriminate global receptive field of deep attention layers introduces contextual interference, causing feature entanglement between adjacent individuals. To mitigate this, Kellnhofer et al. [19] proposed geometric gaze fields, while Dai et al. [6] explored multi-scale fusion for omnidirectional gaze estimation. However, existing approaches either lack explicit geometric isolation or employ rigid geometric assumptions without scene adaptability. Consequently, implicit attention alone remains insufficient for reliable gaze localization in complex, cluttered environments.

3 Method

3.1 Overall Architecture

Given an input image $I \in \mathbb{R}^{3 \times H_I \times W_I}$ containing a target person and their corresponding head bounding box $\mathbf{b} = [x_{min}, y_{min}, x_{max}, y_{max}]$, our objective is to predict a probability heatmap $\mathbf{M} \in [0, 1]^{H_o \times W_o}$, where the position of the maximum value represents the final fixation point of the person's gaze.

As illustrated in Figure 3, we propose **GazeSpot**, a highly efficient single-branch network designed to break the representation bottleneck imposed by single-layer features of foundation models. The overall architecture consists of three core stages:

(1) **Foundation Feature Extraction:** We employ a pre-trained vision foundation model (e.g., DINOv3) as a frozen feature encoder. Unlike previous methods that exploit only the deepest features, we explicitly extract K intermediate features uniformly distributed across different depths of the network to construct a multi-scale feature hierarchy $\mathcal{F} = \{\mathbf{F}_k\}_{k=1}^K$ that encompasses both fine-grained geometry and global semantics.

(2) **Geometric and Semantic Routing:** This is our core contribution. First, the **Geometry-Guided Spatial Focus (GGSF)** module leverages the physical coordinates of the head to generate an explicit viewing frustum mask, performing spatial filtering on the multi-layer features $\mathcal{F}$ to proactively filter out irrelevant backgrounds. Subsequently, the **Scale-Aware Semantic Aggregation (SASA)** module acts as a cognitive engine. It adaptively assigns attention weights to features of different scales based on the global context and fuses them along the channel dimension into a highly condensed scene representation.

(3) **Gaze Decoding and Out-of-Frame Prediction:** The fused representation, combined with positional encodings and the head position prior, is fed into a lightweight Transformer decoder alongside a dedicated state token. The network ultimately outputs a high-resolution gaze heatmap and the classification probability of whether the target is out-of-frame in parallel.

3.2 Geometry-Guided Spatial Focus (GGSF)

Deep vision networks often lose relative spatial position information during the encoding process, leading to a diffuse activation distribution when predicting gaze directions. To address this pain point, we abandon the conventional practice of implicitly learning spatial positions by stacking convolutions. Instead, we propose to introduce an explicit geometric prior via the GGSF module.

The core intuition behind GGSF is to construct a “spotlight-like” attention mechanism for the network. Specifically, for any spatial location $\mathbf{p} = (x, y)$ on the feature map, we first calculate its relative coordinates with respect to the person's head center $\mathbf{c} = (c_x, c_y)$. To endow this coordinate field with scale-aware capability, we further concatenate the absolute physical dimensions of the head (width w and height h). Consequently, we construct a 4-channel geometric tensor $\mathbf{G} \in \mathbb{R}^{4 \times H \times W}$ as follows:

$$\mathbf{G}(\mathbf{p}) = [x - c_x, y - c_y, w, h] \quad (1)$$

Subsequently, we feed $\mathbf{G}$ into a lightweight Multi-Layer Perceptron (MLP), consisting of two 1×1 convolutional layers, Batch-Norm, and ReLU activations, to generate a spatial gating mask

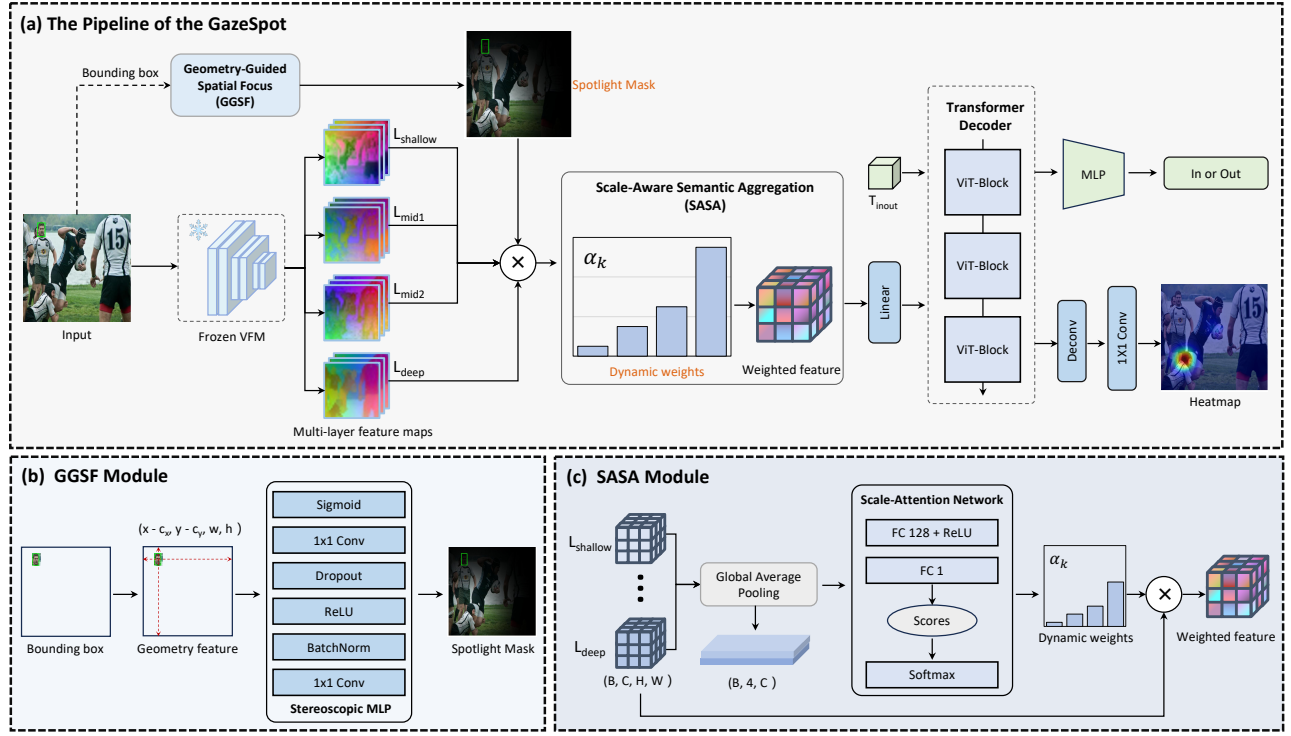


Figure 3: Illustration of GazeSpot. (a) illustrates the overall pipeline. The input image is fed into a frozen VFM to extract multi-layer foundation features. These features first interact with the explicit frustum prior through the Geometry-Guided Spatial Focus (GGSF) module to proactively filter out background noise. Then, the Scale-Aware Semantic Aggregation (SASA) module dynamically allocates attention weights to adaptively fuse the spatially-filtered features across different scales. Finally, the fused representations are decoded by a lightweight Transformer to predict the gaze heatmap and out-of-frame probability. (b) shows the details of the GGSF module, which generates a Spotlight Mask via a stereoscopic MLP based on a relative coordinate field. (c) shows the details of the SASA module, which acts as a cognitive engine for cross-scale feature routing.

$\mathbf{M}_{geo} \in \mathbb{R}^{1 \times H \times W}$ with values ranging in $[0, 1]$:

$$\mathbf{M}_{geo} = \sigma(\text{MLP}(\mathbf{G})) \quad (2)$$

where σ denotes the Sigmoid activation function. Notably, we scale up the bias initialization of the last MLP layer (e.g., set to 2.0) to ensure that the mask values are close to 1 during the early stages of training, thereby preserving the stability of the pre-trained VFM features.

Finally, by employing a multiplicative gating mechanism, the spatial mask $\mathbf{M}_{geo}$ is applied element-wise to the feature maps $\mathbf{F}_k$ at each scale, as described in Equation 3, thereby proactively erasing irrelevant background noise outside the “gaze cone” during the early stages of feature fusion.

$$\mathbf{F}_k^{geo} = \mathbf{F}_k \otimes \mathbf{M}_{geo}, \quad \forall k \in \{1, \dots, K\} \quad (3)$$

This explicit spatial constraint significantly reduces the search space for subsequent semantic optimization.

3.3 Scale-Aware Semantic Aggregation (SASA)

After the spatial filtering by GGSF, the model must determine which layer of features to rely on more heavily for the current gaze scenario. For instance, in a densely crowded scene, distinguishing

the specific target person’s gaze direction from adjacent individuals heavily relies on the sharp geometric boundaries provided by shallow layers, whereas understanding the overarching social interaction or the salient distal object being looked at demands the global semantic context from deep layers. To this end, we propose the SASA module, endowing the model with cross-scale routing capabilities.

Given the geometry-gated multi-layer feature set $\{\mathbf{F}_k^{geo}\}_{k=1}^K$, where $\mathbf{F}_k^{geo} \in \mathbb{R}^{C \times H \times W}$, we first extract a global semantic descriptor $\mathbf{v}_k \in \mathbb{R}^C$ for each layer via Global Average Pooling (GAP):

$$\mathbf{v}_k = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{F}_k^{geo}(i, j) \quad (4)$$

Next, we utilize a shared scale-attention network to evaluate the relative importance of each descriptor for the current scene, and normalize them across the scale dimension via a Softmax function to obtain dynamic weights α_k :

$$\alpha_k = \frac{\exp(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{v}_k))}{\sum_{j=1}^K \exp(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{v}_j))} \quad (5)$$

where W_1 and W_2 represent the weights of fully connected layers, and δ denotes the ReLU activation function.

Upon obtaining the weights, SASA adaptively weights the features and concatenates them along the channel dimension. A 1×1 convolution is then applied to reduce the dimension to a unified hidden dimension D ($D = 256$ in this paper), yielding the final aggregated feature $X \in \mathbb{R}^{D \times H \times W}$:

$$X = \text{Conv}_{1 \times 1} \left(\text{Concat}_{k=1}^K (\alpha_k \cdot F_k^{\text{geo}}) \right) \quad (6)$$

The ingenuity of SASA lies in liberating the network from the representation bottleneck of any single depth. Instead, it transforms the network into a flexible system capable of dynamically allocating the “geometric-semantic” synergy based on the image content.

3.4 Gaze Decoding and Out-of-Frame Prediction

In real-world video interaction scenarios (e.g., the VideoAttention-Target dataset), a person’s gaze target often lies out-of-frame. To construct a comprehensive gaze estimation system, our decoder must not only predict in-frame gaze heatmaps but also possess the binary classification capability to determine whether the target is within the frame.

Given the 2D feature X fused by SASA, we first flatten its spatial dimensions and transpose it into a 1D feature sequence $S \in \mathbb{R}^{(HW) \times D}$. To preserve spatial position information and the person identity prior, we perform an element-wise addition of the sequence with a learnable 2D positional encoding E_{pos} [8] and a head map embedding E_{head} :

$$S' = S + E_{\text{pos}} + E_{\text{head}} \quad (7)$$

where E_{head} is obtained by mapping the person’s original bounding box region into a mask and multiplying it by a learnable head token. This is designed to explicitly guide the model to focus on the specific person currently under test.

To enable out-of-frame gaze prediction, we introduce a dedicated learnable token $T_{\text{inout}} \in \mathbb{R}^{1 \times D}$, concatenated at the forefront of the sequence:

$$S_{\text{input}} = [T_{\text{inout}}, S'] \quad (8)$$

Subsequently, the complete sequence S_{input} is fed into an encoder consisting of L ($L = 3$ in this paper) standard Transformer blocks [38]. Leveraging the self-attention mechanism, the model can perform deep, cross-domain information interaction between head features and scene context within a global receptive field.

The output sequence of the Transformer is denoted as $[\hat{T}_{\text{inout}}, \hat{S}]$. We separate it to perform two parallel tasks:

(1) Out-of-Frame Prediction: The initial token $\hat{T}_{\text{inout}}$ is fed into a classification head composed of an MLP to predict the probability $\hat{p}_{\text{inout}}$ that the fixation point is located within the frame:

$$\hat{p}_{\text{inout}} = \sigma(\text{MLP}(\hat{T}_{\text{inout}})) \quad (9)$$

(2) Gaze Heatmap Generation: The remaining sequence $\hat{S}$ is reshaped back into a 2D feature map and fed into a heatmap decoding head (comprising deconvolutional upsampling and a 1×1 convolution). This ultimately outputs a high-resolution gaze probability heatmap $\hat{M} \in [0, 1]^{H_o \times W_o}$:

$$\hat{M} = \sigma(\text{Deconv}(\text{Reshape}(\hat{S}))) \quad (10)$$

3.5 Optimization Objective

During the end-to-end training process, we unify the heatmap prediction and out-of-frame gaze prediction into pixel-level and image-level Binary Cross-Entropy (BCE) optimization problems.

For in-frame gaze heatmap prediction, since the fixation point is typically modeled as a Gaussian distributed region, we employ the BCE Loss to pixel-wisely constrain the difference between the predicted heatmap $\hat{M}$ and the ground-truth Gaussian heatmap M_{gt} :

$$\mathcal{L}_{\text{heat}} = -\frac{1}{H_o W_o} \sum_{i,j} [M_{gt}^{i,j} \log \hat{M}^{i,j} + (1 - M_{gt}^{i,j}) \log(1 - \hat{M}^{i,j})] \quad (11)$$

For out-of-frame gaze classification, we similarly use the BCE Loss to constrain the predicted probability $\hat{p}_{\text{inout}}$ against the ground-truth binary label $p_{gt} \in \{0, 1\}$ (1 indicates in-frame, 0 indicates out-of-frame):

$$\mathcal{L}_{\text{inout}} = -[p_{gt} \log \hat{p}_{\text{inout}} + (1 - p_{gt}) \log(1 - \hat{p}_{\text{inout}})] \quad (12)$$

The final overall optimization objective function $\mathcal{L}_{\text{total}}$ is the weighted sum of the two, as described in Equation 13, where λ is a hyperparameter balancing the two tasks.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{heat}} + \lambda \mathcal{L}_{\text{inout}} \quad (13)$$

Specifically, on datasets containing only in-frame gaze (e.g., Gaze-Follow), λ is set to 0; whereas on datasets containing out-of-frame gaze (e.g., VideoAttentionTarget), this term is activated.

4 Experiments

4.1 Datasets and Evaluation Metrics

To comprehensively evaluate the performance and robustness of GazeSpot, we conduct extensive experiments on two highly challenging and authoritative large-scale public datasets, and specifically construct an evaluation subset tailored for crowded scenes.

Table 1: Sample distribution of the Crowd Subset across different density levels.

Scene Density (BBboxes)	Number of Samples (Frames)
< 3 (Sparse)	7,945
3 (Moderate)	3,015
4 (Crowded)	1,628
> 4 (Extremely Crowded)	539

GazeFollow [27]: This dataset is the most widely used benchmark in the field of gaze target estimation, comprising a massive number of static images in real-world scenarios. Since the same person in an image may correspond to different fixation points annotated by multiple annotators, this dataset primarily evaluates the model’s spatial inference capabilities in complex scenes.

VideoAttentionTarget (VAT) [5]: This is a large-scale, highly dynamic video dataset. Unlike GazeFollow, VAT not only contains complex social interactions but also introduces “out-of-frame” gaze annotations, requiring the model to possess the binary classification capability to determine whether the target falls outside the image boundaries.

Table 2: Comparison with State-of-the-Art methods on GazeFollow and VideoAttentionTarget datasets. The best results are highlighted in bold.

Category	Methods	Venue	Params	GazeFollow			VideoAttentionTarget		
				AUC $\uparrow$	Avg $L_2 \downarrow$	Min $L_2 \downarrow$	AUC $\uparrow$	$L_2 \downarrow$	In/Out AP $\uparrow$
Multi-branch / Multi-modal	Chong et al. [4]	ECCV 2018	51M	0.896	0.187	0.112	0.833	0.171	0.712
	Lian et al. [20]	ACCV 2018	55M	0.906	0.145	0.081	0.837	0.165	-
	Chong et al. [5]	CVPR 2020	61M	0.921	0.137	0.077	0.860	0.134	0.853
	Fang et al. [10]	CVPR 2021	68M	0.922	0.124	0.067	0.905	0.108	0.896
	Bao et al. [1]	CVPR 2022	29M	0.928	0.122	-	0.885	0.120	0.869
	Jin et al. [18]	EAAI 2022	52M	0.920	0.118	0.063	0.900	0.104	0.895
	Miao et al. [24]	WACV 2023	61M	0.934	0.123	0.065	0.917	0.109	<u>0.908</u>
	Tafasca et al. [34]	ICCV 2023	25M	0.939	0.122	0.062	0.914	0.109	0.834
VFM-based Single-branch	Tafasca et al. [35]	CVPR 2024	105M	0.944	0.113	0.057	-	0.107	0.891
	ViTGaze [33]	VI 2024	22M	0.949	0.105	0.047	0.938	0.102	0.905
	GT360 [6]	IROS 2025	1.9M	0.957	0.101	-	0.934	0.103	0.887
	Gaze-LLE(ViT-B) [29]	CVPR 2025	2.8M	0.956	0.104	0.045	0.933	0.107	0.897
Ours	Gaze-LLE(ViT-L) [29]	CVPR 2025	2.9M	<u>0.958</u>	<u>0.099</u>	<u>0.041</u>	0.937	0.103	0.903
	GazeSpot (ViT-B)	-	3.5M	0.958	0.101	0.043	0.940	0.099	0.887
Ours	GazeSpot (ViT-L)	-	3.8M	0.961	0.093	0.038	0.944	0.092	0.912

Crowd Subset: To precisely quantify the model’s robustness against contextual interference, we conduct a fine-grained reconstruction of the test set from the VAT dataset. Specifically, we strictly categorize the test samples into four density levels based on the number of ground truth head bounding boxes in each frame: < 3 (Sparse), 3 (Moderate), 4 (Crowded), and > 4 (Extremely Crowded). This rigorous quantitative division allows us to clearly track the performance degradation trajectory of models as the number of distractors increases. The specific frame distribution across different density levels is detailed in Table 1.

Evaluation Metrics: Regarding evaluation metrics, we strictly adhere to the standard protocols of this field. For in-frame gaze, we employ AUC (Area Under Curve) by treating the heatmap as classification confidences to evaluate overall prediction accuracy. Simultaneously, we use the L_2 Error to calculate the Euclidean distance between the predicted hottest point and the ground-truth fixation point (reporting Average L_2 and Minimum L_2 for GazeFollow). For out-of-frame gaze (exclusive to the VAT dataset), we employ AP (Average Precision) to evaluate the average precision of the binary classification.

4.2 Implementation Details

Our model is implemented using the PyTorch framework, and all experiments are conducted on a single NVIDIA RTX 3090 GPU, demonstrating exceptional computational efficiency. We adopt the officially pre-trained DINOv3 as our vision foundation model and strictly freeze its weights throughout the entire training process. The input images are uniformly resized to a resolution of 512×512 , and the model ultimately outputs a gaze heatmap with a resolution of 64×64 . Within the core modules of GazeSpot, the hidden dimension is set to $D = 256$, and the number of Transformer layers in the decoder is set to $L = 3$.

For the optimization strategy, we employ the Adam optimizer [7] combined with a Cosine Annealing [22] learning rate schedule. For the VAT dataset, due to the inherent difficulty disparity between out-of-frame gaze prediction and heatmap regression, we adopt asymmetric learning rates: the learning rate for the classification head responsible for in/out prediction is set to 10^{-3} , while the rest of the network is set to 10^{-5} . Both heatmap regression and out-of-frame classification utilize the Binary Cross-Entropy (BCE) loss, with the multi-task balancing weight λ set to 1.0. More details and code are provided in the Supplementary Materials.

4.3 Comparison with State-of-the-Art

Results on Public Benchmarks. We comprehensively compare GazeSpot with recent top-tier methods on both the GazeFollow and VideoAttentionTarget benchmarks, as detailed in Table 2. We categorize the comparative methods into traditional “multi-branch/multi-modal architectures” and emerging “VFM-based single-branch architectures.” Experimental results demonstrate that GazeSpot achieves new SOTA performance. Specifically, our large variant, GazeSpot (ViT-L), establishes absolute SOTA records with 0.961 AUC and 0.093 L_2 error on GazeFollow, as well as 0.944 AUC and 0.092 L_2 error on VideoAttentionTarget. Most notably, even our base variant, GazeSpot (ViT-B), significantly outperforms the parameter-heavy Gaze-LLE (ViT-L) on the highly dynamic VideoAttentionTarget dataset (0.940 vs. 0.937 AUC, 0.099 vs. 0.103 L_2). This strongly proves that the explicit geometric and multi-scale routing mechanisms introduced through SASA and GGSF possess indisputable superiority on general benchmarks, enabling our streamlined architecture to surpass previous large-scale models.

Results in Crowded Scenes. To expose the vulnerability of traditional single-branch vision foundation model in complex social scenarios, we conduct a fine-grained degradation trajectory analysis on the constructed Crowd Subset, with quantitative results

Table 3: Performance comparison and degradation analysis across different crowd densities.

Methods	Metrics	< 3	3	4	> 4
Gaze-LLE [29]	AUC $\uparrow$	0.958	0.932	0.934	0.875
	L_2 $\downarrow$	0.078	0.100	0.111	0.182
	In/Out AP $\uparrow$	0.916	0.811	0.928	0.944
GazeSpot	AUC $\uparrow$	0.958	0.934	0.942	0.890
	L_2 $\downarrow$	0.077	0.099	0.101	0.171
	In/Out AP $\uparrow$	0.923	0.798	0.916	0.967
Relative L_2 Reduction		1.3%	1.0%	9.0%	6.0%

detailed in Table 3. As the number of individuals in the frame increases—indicating intensified contextual interference—both models naturally experience performance degradation due to the escalating difficulty of the task. However, the baseline model (Gaze-LLE) suffers a much steeper degradation trajectory. While GazeSpot and the baseline perform comparably in sparse scenes (< 3), the performance gap between them steadily widens as the crowd density increases. Specifically, the L_2 error reduction achieved by GazeSpot over the baseline expands from a marginal 0.0015 in moderate scenes (=3) to 0.0092 in crowded scenes (=4), and ultimately reaches a significant 0.0109 in extremely crowded environments (> 4), achieving a relative error reduction of nearly 6% (0.171 vs. 0.182). Benefiting from the frustum physical isolation of GGSF and the multi-scale routing of SASA, GazeSpot’s growing relative advantage in denser crowds perfectly validates the theoretical expectation that multiplicative gating effectively cuts off attention hijacking.

Qualitative Evaluation. To intuitively demonstrate the advantages in anti-interference capability, Figure 4 provides a qualitative visual comparison between the baseline model and GazeSpot on the Crowd Subset. Clearly, in complex scenes with dense crowd interference, the baseline model’s attention is highly prone to diffusion. It is easily distracted by irrelevant background individuals possessing strong semantics, causing the predicted heatmap to deviate from the ground-truth fixation point. Conversely, relying on explicit geometric gating, GazeSpot effectively filters out background noise outside the viewing frustum, as illustrated by the Spotlight Mask in the figure. Even when facing interference from crowd occlusions, it still achieves highly compact and physically aligned spatial focus. This intuitively confirms the effectiveness of our proposed method in alleviating contextual interference.

Model Complexity and Efficiency. While achieving exceptional robustness, GazeSpot maintains ultimate lightweight efficiency. Traditional multi-branch models easily incur parameter counts up to 50M or even over 100M. In contrast, since we completely freeze the foundation model, our GazeSpot possesses a mere 3.5M learnable parameters. As shown in Table 2, compared to the baseline model Gazelle (2.8M), we only introduce a minuscule parameter overhead of 0.7M, primarily originating from the extremely lightweight SASA routing network and the GGSF gating MLP. This compellingly proves that the massive performance leap of GazeSpot in Crowd scenes stems from a profound physical mechanism reconstruction, rather than brute-force parameter stacking.



Figure 4: Qualitative visual comparison between the baseline model and GazeSpot on the Crowd Subset. From top to bottom: input image with the target person’s head annotated by a green bounding box, ground-truth fixation point annotation, frustum physical isolation mask generated by GazeSpot (Spotlight Mask), gaze heatmap predicted by the baseline model, and gaze heatmap predicted by our GazeSpot.

4.4 Ablation Study

To validate the effectiveness of each core component, we conduct rigorous ablation studies on GazeFollow and VideoAttentionTarget. Results are presented in Table 4.

Baseline (Last-layer only). Using only the deepest VFM features achieves 0.955 AUC and 0.107 L_2 error on GazeFollow. This validates our motivation: even with DINOv3’s strong semantic capacity, single-layer representations hit a performance ceiling due to geometric detail loss.

Effect of Multi-layer Fusion. Simply concatenating multi-layer features without guidance (row 2) yields marginal and inconsistent gains, specifically +0.002 AUC on GazeFollow and +0.004 AUC on VideoAttentionTarget. We attribute this to low-frequency noise from shallow layers diluting the semantic signal because brute-force fusion lacks the adaptive routing required to resolve the spatial-semantic dilemma.

Effect of SASA. Adding the Scale-Aware Semantic Aggregation module (row 3) unlocks the potential of multi-layer features, improving GazeFollow AUC to 0.958 (+0.003 over raw concatenation) and reducing L_2 error to 0.102. The dynamic routing mechanism successfully extracts occlusion-resistant shallow edges when needed, as evidenced by the 9.0% relative L_2 reduction (as shown in Table 3) in crowded scenes.

Effect of GGSF. Integrating the Geometry-Guided Spatial Focus atop SASA (row 4) constitutes the complete GazeSpot. This yields our final performance: 0.958 AUC and 0.101 L_2 on GazeFollow, 0.940 AUC and 0.099 L_2 on VideoAttentionTarget. The frustum isolation mechanism eliminates divergent activations behind the target, rendering heatmaps compact and physically aligned (Figure 4).

Table 4: Ablation study evaluating the individual contributions of Multi-layer Features, SASA, and GGSF on the GazeFollow and VideoAttentionTarget datasets (Backbone: DINOv3 ViT-B/16). The best results are highlighted in bold.

	Multi-layer	SASA	GGSF	GazeFollow		VideoAttentionTarget	
				AUC $\uparrow$	L_2 $\downarrow$	AUC $\uparrow$	L_2 $\downarrow$
				0.955	0.107	0.933	0.106
	✓			0.957	0.103	0.937	0.103
	✓	✓		0.958	0.102	0.939	0.101
	✓	✓	✓	0.958	0.101	0.940	0.099

4.5 Analysis of Feature Over-smoothing

To deeply corroborate the hypothesis of “feature over-smoothing in large models” proposed in our [section 1](#), we conduct a feature-level analysis across different versions of DINO under the baseline architecture. The quantitative comparison is detailed in [Table 5](#).

A counterintuitive phenomenon is observed: when utilizing a pure single-branch baseline architecture (extracting only the last-layer features), upgrading the backbone from DINOv2 to DINOv3 not only fails to bring performance improvements in gaze estimation but actually enlarges the L_2 error. The Principal Component Analysis (PCA) visualization (as shown in [Figure 2](#)) reveals its underlying pathology: DINOv3 possesses formidable global semantic homogeneity, yet exhibits severe feature over-smoothing in densely populated local regions. This over-smoothing erases the geometric boundaries between distinct individuals, causing the model to plunge into severe semantic confusion during prediction.

However, this phenomenon does not imply that DINOv3 is unsuitable. Precisely by leveraging the SASA routing and GGSF geometric gating mechanisms proposed by GazeSpot, we successfully leap over the resolution trap of DINOv3’s last layer. By non-intrusively extracting its uncontaminated, early high-frequency features, we completely unleash the unfathomable representation potential of DINOv3. Experimental results show that DINOv3 equipped with the GazeSpot architecture achieves an overwhelming reversal, perfectly closing the loop on this paper’s core argument regarding “reconstructing the geometric-semantic balance” of foundation models.

5 Limitation

Despite the effectiveness of the proposed approach, several limitations remain. The integration of multi-level feature aggregation inevitably introduces additional computational cost compared to lightweight single-layer designs such as Gaze-LLE, which may pose challenges for real-time deployment or resource-constrained scenarios. In addition, the geometry-guided spatial constraint relies on the quality of head-related cues to construct the spatial prior. When head pose estimation becomes unreliable, particularly in cases of heavy occlusion or extreme viewpoints (as shown in [Figure 5](#)), the resulting spatial constraint may be less accurate, potentially affecting overall performance. Future work may explore more efficient feature aggregation strategies, as well as adaptive spatial modeling mechanisms that can better handle uncertain or noisy geometric cues.

Table 5: Performance comparison demonstrating the over-smoothing effect of DINOv3 in the baseline architecture, and the subsequent overwhelming reversal when integrated with the GazeSpot framework. All metrics report the L_2 error ($\downarrow$).

Dataset	Baseline (Gazelle)		GazeSpot (Ours)	
	DINOv2	DINOv3	DINOv2	DINOv3
GazeFollow	0.104	0.107	<u>0.102</u>	0.101
VideoAttentionTarget	0.107	0.106	<u>0.104</u>	0.099
Crowd (≥ 4)	0.133	0.139	<u>0.132</u>	0.124



Figure 5: Visual examples of failure cases. In scenarios involving heavy head occlusion or extreme viewpoints, the spatial prior (Spotlight Mask) becomes unreliable, subsequently leading to inaccurate or diffuse gaze target predictions.

6 Conclusion

In this paper, we revisit gaze target estimation from the perspective of both representation and spatial reasoning. While recent approaches based on vision foundation models have achieved strong performance using simplified architectures, they largely rely on single-layer representations and implicit attention mechanisms, which may limit their effectiveness in complex scenes. We show that accurate gaze estimation requires jointly modeling fine-grained spatial structures and high-level semantics. To this end, we propose a unified framework that integrates multi-level feature aggregation with geometry-guided spatial constraints. By leveraging complementary information across feature hierarchies and explicitly constraining the attention space, our approach improves robustness against contextual interference. Extensive experiments on challenging benchmarks demonstrate that the proposed method consistently outperforms existing approaches, particularly in crowded scenarios. Beyond gaze estimation, our findings suggest that combining multi-level representations with explicit spatial priors is a promising direction for improving dense prediction tasks based on vision foundation models.

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